HAW Hamburg

Fachbereich Elektrotechnik und Informatik

Semester: [SS] **2025** **Lab group (DCL 02)**	**Performed tasks:** Hands-on hardware	**Protocol manager:** Sajjadur Rahman
Team name/number:	**Date of exercise:** 10.07.2025	**Other participants:** Ananna Saha Minhazul Islam Meherun Afroz Nabila Mohammad Shadikur Rahman
Professor: Rainer Schoenen	**Attestation:**	
Digital Communications – Experiment DCL-4		**SHN/12.05.2025**

I/Q Complex Modulation, wireless

1 Introduction

The purpose of this lab was to investigate and understand channel coding, modulation, demodulation, and decoding, with a focus on coherent I/Q modulation schemes such as QPSK and QAM variants. Both wired and wireless scenarios were examined to study practical performance, signal degradation, and visualization through constellation and eye diagrams.

Modern communication systems rely heavily on advanced modulation schemes to increase spectral efficiency and achieve higher data rates. Understanding these schemes and their behavior in realistic channels is crucial for designing reliable and robust systems.

2 Theory

Digital modulation refers to mapping digital information onto an analog carrier signal for transmission. The core idea is to vary certain properties of a carrier waveform (amplitude, phase, or frequency) to represent binary data.

Quadrature Phase Shift Keying (QPSK)

QPSK is a phase modulation technique that uses four different phase shifts (0°, 90°, 180°, and 270°), allowing two bits per symbol. Compared to BPSK, QPSK effectively doubles the data rate without increasing the bandwidth. In the constellation diagram, four distinct points are observed corresponding to each symbol. QPSK offers good noise immunity and is widely used in wireless and satellite communications.

Quadrature Amplitude Modulation (QAM)

QAM combines amplitude and phase variations to convey data. 16QAM uses 16 constellation points, allowing 4 bits per symbol, while higher orders like 64QAM and 256QAM encode 6 and 8 bits per symbol, respectively.

While higher-order QAM increases data rates, it also makes the system more sensitive to noise and requires higher signal-to-noise ratios (SNR). This trade-off is critical when designing communication links for different channel conditions.

Eye and Constellation Diagrams

The constellation diagram visually represents the signal modulation accuracy. A well-defined, tightly clustered constellation indicates less distortion and lower error rates.

The eye diagram helps analyze the time-domain behavior, providing insights into intersymbol interference, timing jitter, and overall signal quality. An open eye pattern suggests good transmission quality, whereas a closed eye indicates higher probability of errors.

Wireless Challenges

In wireless scenarios, additional factors such as multipath fading, Doppler effects, and antenna alignment impact signal quality. Imperfect synchronization and environmental

obstacles lead to constellation spreading and eye closure, which degrade system performance.

3 Objectives

- Understand and implement I/Q coherent modulation.
- Generate and analyze QPSK, 16QAM, 64QAM, and 256QAM signals.
- Visualize modulation performance using constellation and eye diagrams.
- Evaluate differences between wired and wireless transmissions.
- Develop deeper insight into the practical limitations and robustness of each scheme.

4 I/Q Coherent Modulation

A Rohde & Schwarz SMIQ3B signal generator was used to produce a 70 MHz RF signal, modulated with different data rates and schemes. Initially, QPSK modulation was generated at 1 kbit/s and later increased to 10 kbit/s to observe changes in signal clarity.

QPSK Constellation

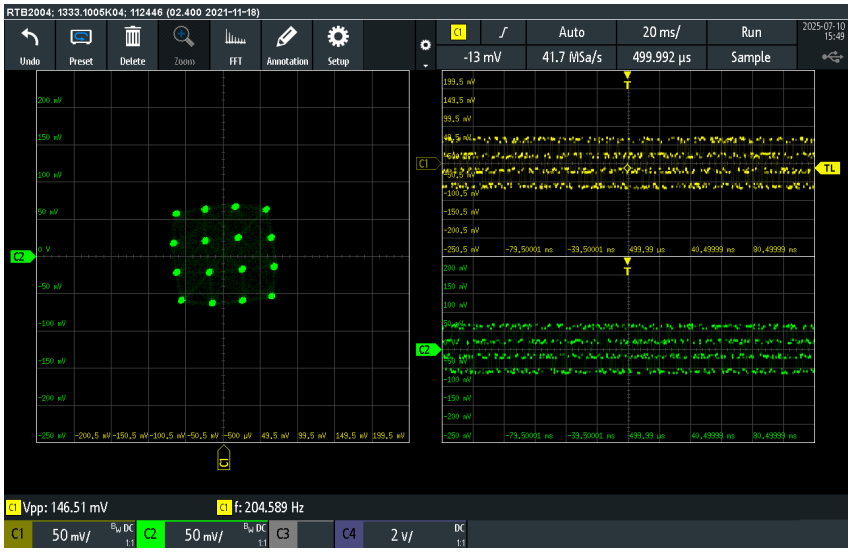


Figure 1: QPSK constellation diagram showing four clear points.

QPSK Eye Diagram (Antenna)

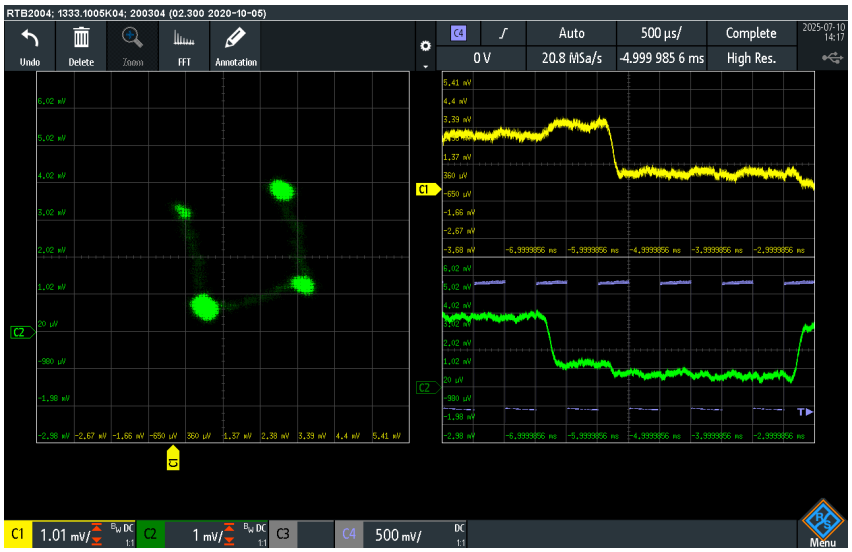


Figure 2: QPSK eye diagram illustrating good symbol separation (antenna setup).

16QAM Constellation

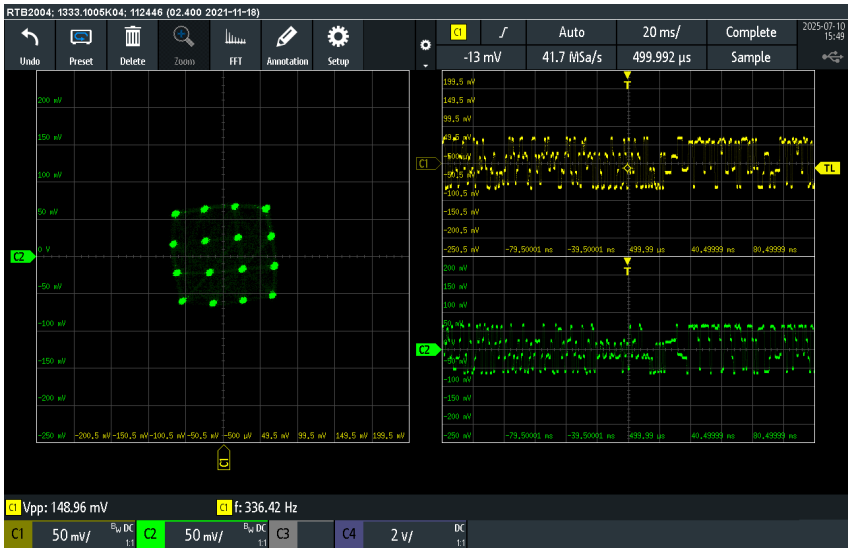


Figure 3: 16QAM constellation diagram with 16 symbol points (wired).

16QAM Eye Diagrams

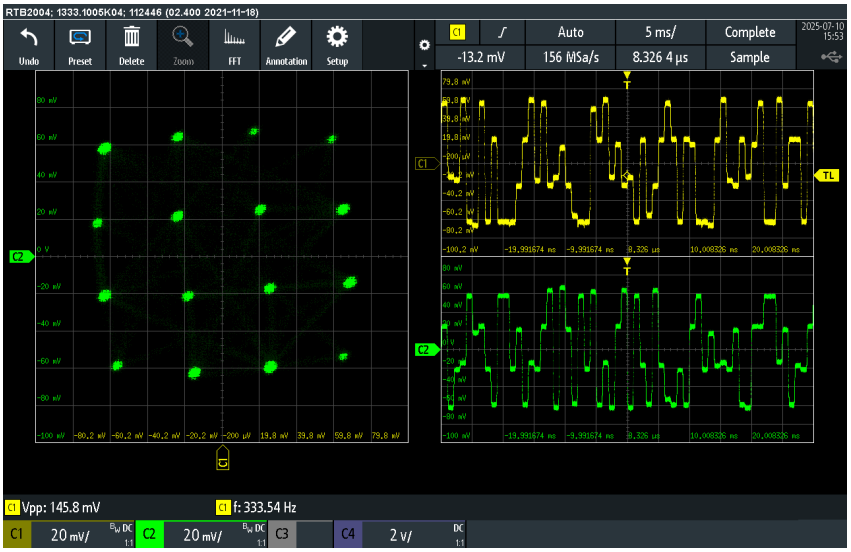


Figure 4: 16QAM eye diagrams for I and Q components (detail 1).

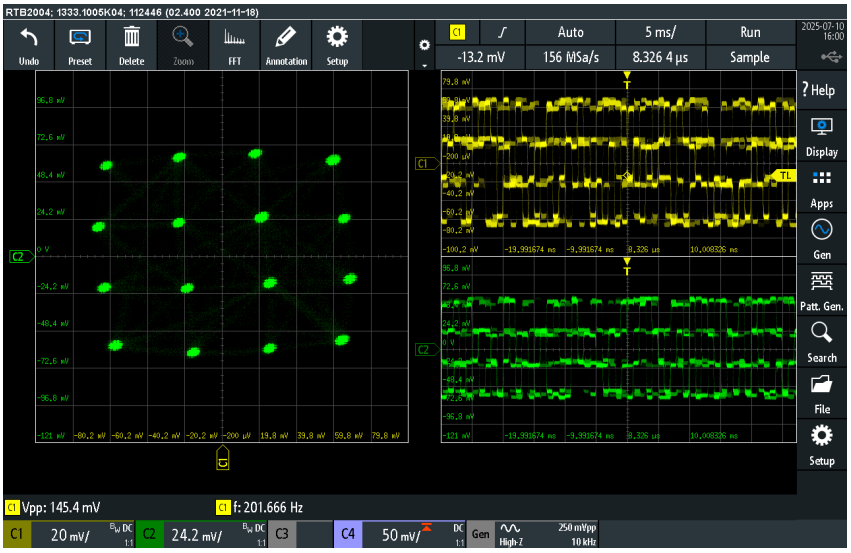


Figure 5: 16QAM eye diagrams for I and Q components (detail 2).

64QAM and 256QAM

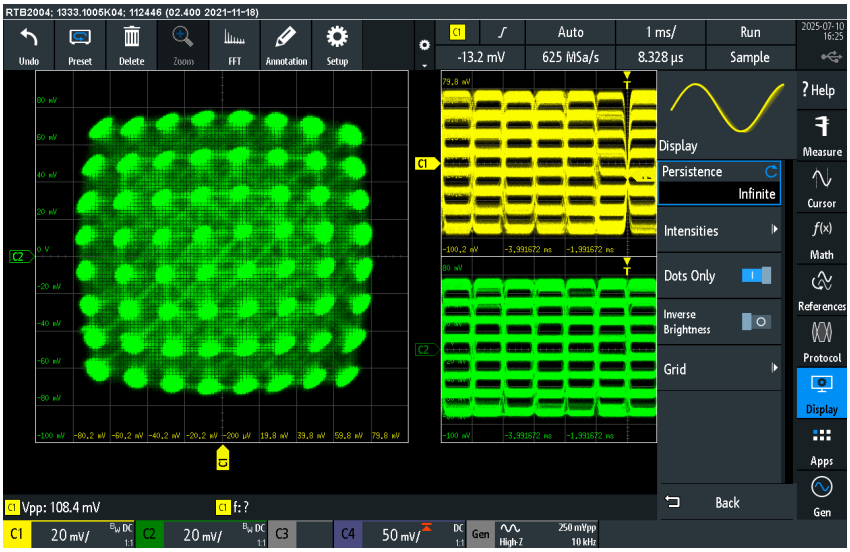


Figure 6: 64QAM constellation diagram with higher symbol density (wired).

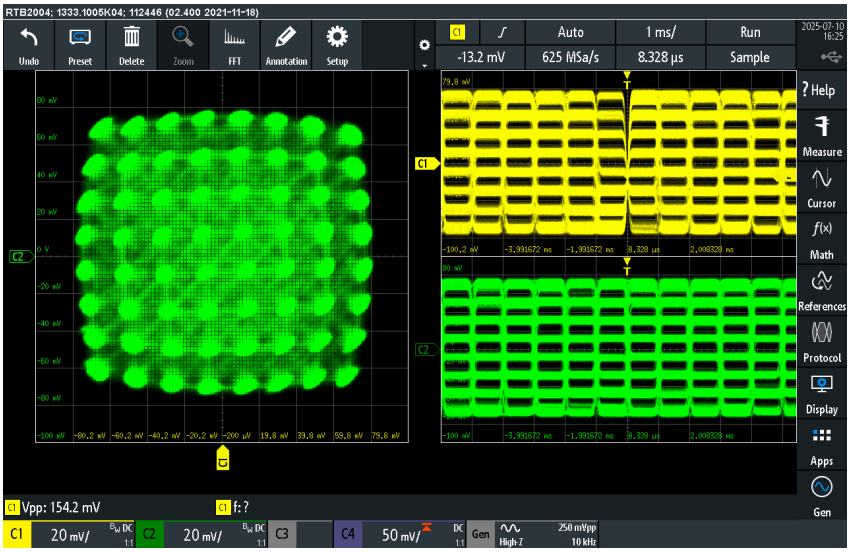


Figure 7: 256QAM constellation diagram showing very high symbol density and greater sensitivity to noise (wired).

5 Wireless Transmission Observations

In the wireless experiments, antennas were used for both transmitting and receiving the RF signal. Various modulation schemes were tested to observe the effects of real-world propagation, including alignment errors and multipath fading.

QPSK with Antenna

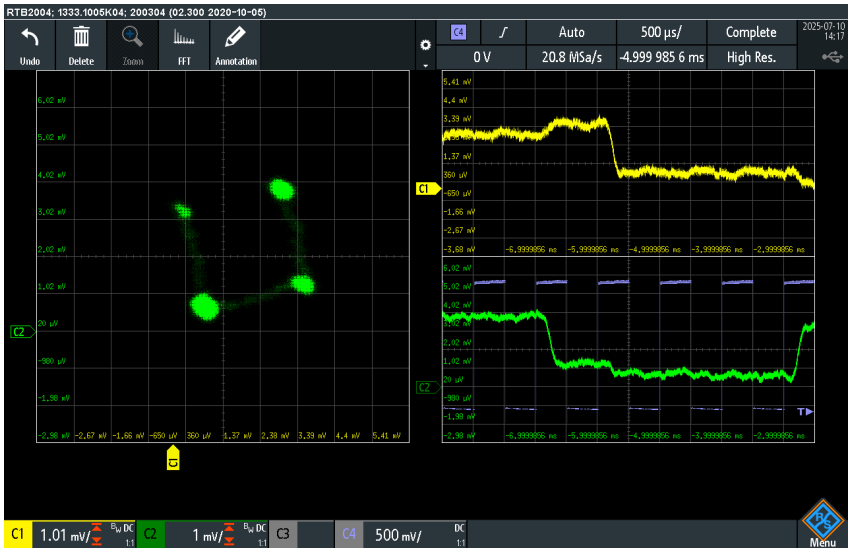


Figure 8: Constellation and eye diagram of QPSK with antenna.

16QAM with Antenna

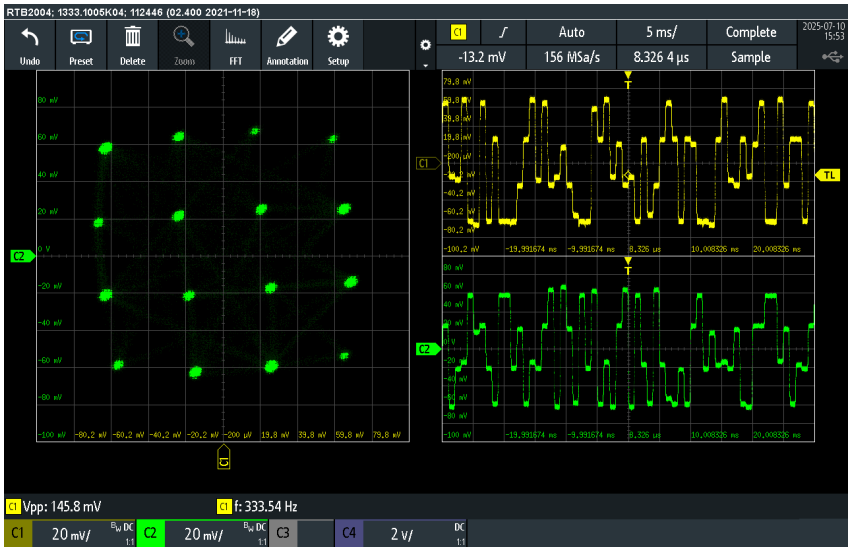


Figure 9: 16QAM constellation diagram obtained with antenna transmission.

64QAM with Antenna



Figure 10: 64QAM constellation diagram received via antenna.

256QAM with Antenna

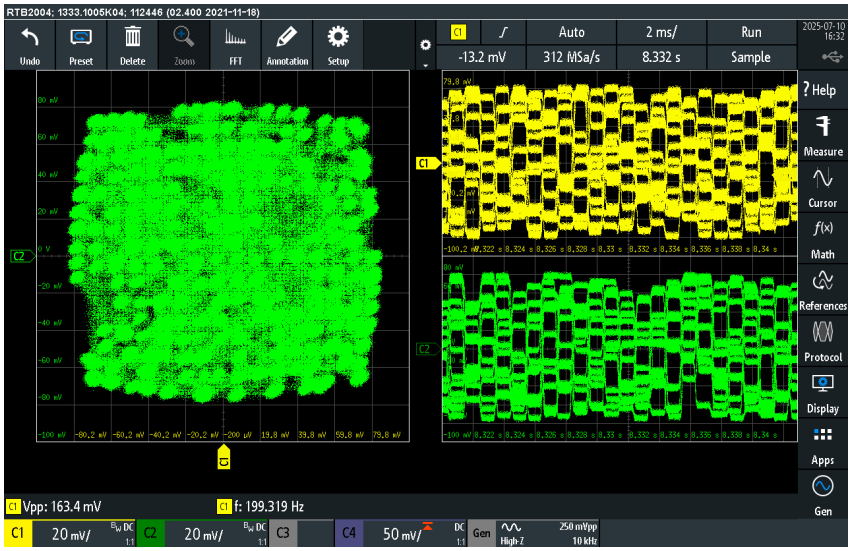


Figure 11: 256QAM constellation diagram via antenna, showing noticeable noise and spreading.

Antenna Blocked Condition

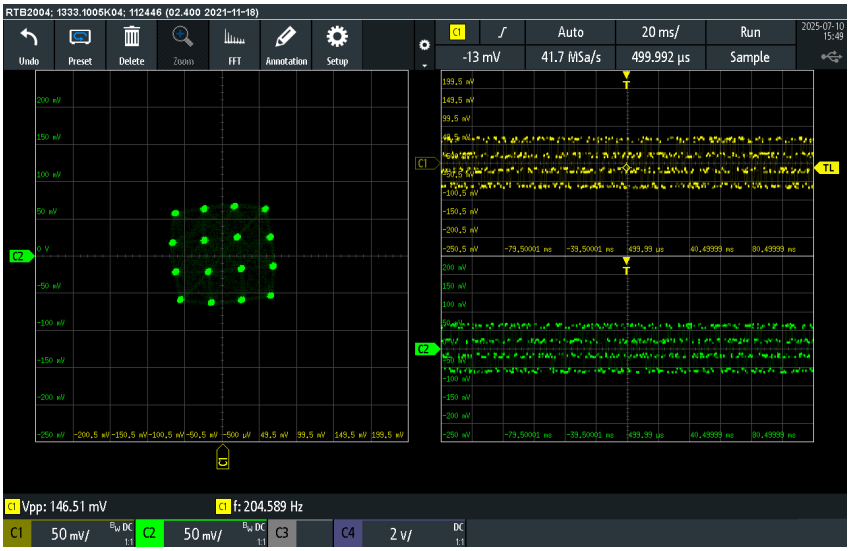


Figure 12: Constellation diagram when antenna is partially blocked, showing distortion and symbol overlap.

6 Discussion and Evaluation

The experiments demonstrated how different modulation schemes behave under ideal and practical conditions. QPSK showed excellent robustness and minimal symbol overlap. 16QAM provided higher data throughput but with reduced tolerance to noise. 64QAM and 256QAM achieved the highest spectral efficiencies; however, their performance rapidly degraded with lower SNR or imperfect antenna alignment.

Wireless experiments highlighted the challenges posed by real-world environments. Fading, multipath, and antenna obstruction led to constellation smearing and eye closure, emphasizing the importance of error correction coding and adaptive modulation in practical systems.

7 Conclusion

Through these experiments, we gained valuable practical insights into the principles of digital modulation, the importance of SNR, and real-world transmission challenges. We successfully visualized constellations and eye diagrams for various modulation schemes, both in wired and wireless setups, deepening our understanding of system performance and the trade-offs involved in modern communication systems.